Interpreting benthic Common Era foraminiferal records under the assumption of modern-day circulation

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Key Points:

- We present an inversion method that incorporates age model uncertainty to infer far-field properties from benthic sediment core records (137/140 characters)
- The reconstructed North Atlantic Little Ice Age is characterized by a 0.11-1.2K/kyr cooling (1σ confidence interval) from 1200-1900CE (140/140 characters)
- A long-term cooling in the Nordic Seas is reconstructed, and is found to be approximately two times greater than that of the subpolar gyre (138/140 characters)

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Abstract

Paleoceanographic reconstructions of the past 2,000 years indicate a large-scale cooling in the latter half of the last millennium, but significant conflict on magnitude and timing exist. This conflict is partially due to the inference of surface conditions from sparse, noisy, near-surface marine proxies that can exhibit seasonal biases. Here, we develop a novel methodology for inferring far-field changes from benthic sediment core records via an ocean circulation model. Benthic sediment core records are not subject to high-energy near-surface variability or seasonal biases and record much more stable signals than equivalent planktonic records. We present a possible surface temperature and $\delta^{18}O_{seawater}$ history that fits a dataset of 11 high-resolution benthic $\delta^{18}O_{calcite}$ records. The estimate presented here is within error of instrumental records, and is characterized by a Little Ice Age sea surface temperature cooling of 0.67 ± 0.56 K/kyr, which is similar to, but at the high end, of previous independent estimates obtained from Common Era marine surface records. Our reconstruction is characterized by a larger magnitude millenial-scale cooling in the Nordic Seas (0.9 ± 2.7 K) than in the subpolar gyre (0.5 ± 2.2K), indicative of an Arctic amplification of Little Ice Age cooling.

Plain Language Summary

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1 Introduction

The past 2,000 years, referred to as the Common Era (CE), is a critical period for understanding the response of the Earth's climate to perturbations on long timescales. Climatic perturbations during the CE include decreasing solar irradiance due to the precession of the equinoxes and stratospheric sulfate injection from volcanic eruptions. During the latter half of the last millennium, a climatic anomaly referred to as the Little Ice Age has been identified, characterized by anomalously cool sea surface temperatures (McGregor et al., 2015) as well as surface air temperatures (PAGES 2k Consortium, 2013). Understanding the spatiotemporal extent of the Little Ice Age is a critical step in constraining the response of the climate system to various natural perturbations on timescales longer than that of the instrumental record. Commonly, the Little Ice Age is thought to have been triggered by either decreasing solar irradiance or volcanic activity, and reinforced via oceanic and atmospheric mechanisms. However, the precise reinforcement mechanisms have yet to be conclusively identified due to the sparse, noisy, and biased nature of paleoclimate proxy records.

Modeling studies have suggested that an increase of Arctic Ocean sea-ice as a direct response to decadally-paced volcanic eruptions beginning in 1257-1258CE may have lead to a freshening of the subpolar North Atlantic that weakened convection (Zhong et al., 2010; Miller et al., 2012; Lehner et al., 2013; Schleussner & Feulner, 2013). Weakened convection has been associated with a slowdown of the Atlantic Meridional Overturning Circulation (AMOC), although some modeling studies have suggested that an AMOC slowdown is not necessary to explain the full magnitude of LIA cooling (Moreno-Chamarro et al., 2016). In addition, paleoceanographic proxy records have not identified large-scale circulation anomalies in the pre-industrial CE (Thornalley et al., 2018; Moffa-Sanchez et al., 2015; Rahmstorf et al., 2015). Dynamics related to a weakened subpolar gyre (Moffa-Sánchez, Born, et al., 2014; ?, ?) have also been suggested to be a key reinforcement mechanism, and have been supported through deep-sea coral records (Copard et al., 2012), bivalve resiliency records (Arellano-Nava et al., 2022), and planktonic foraminiferal records (Moffa-Sánchez, Born, et al., 2014; Moffa-Sanchez & Hall, 2017; Spooner et al.,

2020), although exact timing and spatial extent of subpolar gyre variability still remain unclear.

One of the primary limitations of our understanding of LIA dynamics comes from the point-wise nature of paleoceanographic proxy records. Inferring behavior of a large-scale oceanographic feature such as the subpolar gyre typically requires spatially dense and high temporal resolution measurements, which can be difficult to attain for paleo-timescales. Estimates of LIA SST cooling rates as well as timing differ amongst reconstructions, from an estimate of 0.2°C/kyr from 1200-1940CE in the northeast North Atlantic (Cunningham et al., 2013), to 0.54°C/kyr from 800-1900CE globally (McGregor et al., 2015). These reconstructions all rely upon surface records, which experience a high degree of noise and seasonal effects. When trying to extract long-term climate signals, benthic records may be a better option, as they are not subject to high-frequency surface variability, nor are they subject to seasonal variability.

In this paper, we develop an inverse approach to inferring far-field properties from benthic proxy records through the usage of a data-informed circulation model. Here, we apply a modern-day circulation model under the assumption that there were no large-scale circulation changes over the CE. Testing whether or not simultaneous sediment core records can be well-fit under the assumption of steady-state modern-day circulation has been previously used as a test for the necessity of circulation change (Burke et al., 2011; Gebbie, 2012; Amrhein et al., 2015). The circulation model used here is a transient inversion of modern-day tracer observations (Gebbie & Huybers, 2012) previously used to constrain CE deep Pacific and Atlantic cooling (Gebbie & Huybers, 2019). This prior application estimated a temperature history that could explain the observed warming from the 1870s HMS Challenger expedition to modern hydrography. It has also been used to explain the lag between Atlantic and Pacific warming in deglacial δ^{18}O records (Gebbie, 2012). Here, we apply this model to CE benthic sediment core δ^{18}O records for the first time. Here, we separate the effects of temperature and $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$ by using experimentally defined ratios (Marchitto et al., 2014).

We will first show the ability of this methodology to fit a novel dataset of 11 high-resolution benthic records of $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ from along the Reykjanes Ridge in Iceland (Lu et al., 2023). Successfully fitting this data implies that possible surface temperature and $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$ histories exist that can explain the sediment core data without necessitating changes in circulation. Next, we will compare our reconstruction to instrumental records of Labrador Sea Water temperature and salinity, and demonstrate that we reconstruct similar magnitudes of temperature and salinity variability over the past 100 years. Our reconstruction can also be compared to other paleoceanographic estimates, and we show that the estimated LIA North Atlantic cooling trend is comparable to the global SST trend of 0.54°C/kyr identified in the Ocean2k dataset. Finally, we will discuss the spatiotemporal evolution of the LIA in the North Atlantic, and quantify the regional magnitudes and timings. In our solution, the Nordic Seas are shown to be the first region to begin cooling, at ∼1200CE, followed by the subpolar gyre in ∼1400CE. The timing of the subpolar gyre cooling is similar to other independent records (Moffa-Sánchez, Hall, et al., 2014). The subpolar gyre is shown to cool by 0.37 ± 2.1K, while the Nordic Seas cool by (0.8 ± 2.7 K). This finding is consistent with the Arctic amplification of the LIA identified by Otto-Bliesner et al. (2016), which was previously suggested to be exhibited in this dataset by larger magnitude $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ variations in deeper sediment cores.

2 Data

2.1 Sediment Core Records from Lu et al. (2023)

The inversion presented here is applied to a set of 11 high-resolution benthic foraminiferal $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ sediment core records (Figure 3) previously presented in Lu et al. (2023). These

records were collected from along the Reykjanes Ridge south of Iceland, and form a depth transect from 1100-2300m. Previously these records were used to demonstrate that the deep ocean LIA transition, as indicated by the benthic records, occurred at 1350 ± 50 CE, while the surface ocean LIA transition, as indicated by planktonic records, occurred from 1100-1700CE. In addition, a larger magnitude LIA was observed in the benthic records than in the planktonic records, indicate of Arctic amplification of LIA temperature changes, transferred to the sediment core sites by an active AMOC (Lu et al., 2023).

Here, we incorporate age model uncertainty into the inverse problem by applying the BACON-generated age models (Blaauw & Christen, 2011; McKay et al., 2021) from Lu et al. (2023). In order to translate the 1000 age model realizations into covariance matrices with uncertainty in the $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ dimension, each age model realization was linearly interpolated at a common 10-year resolution. Then, for each timestep, the mean $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ was computed, for any interpolated timestep that had more than 100 realizations available. The covariance between two arbitrary timesteps t_1, t_2 can be computed according to Equation 1, where $\delta^{18}\mathrm{O}_i(t_1)$ refers to the interpolated $\delta^{18}\mathrm{O}$ value at t_1 for realization i and M is the total number of realizations with data at t_1 and t_2. Realizations will have different start and end times due to the age modeling process, so there are not necessarily 1000 realizations at every timestep.

[NOTE: This may be a good spot for a supplementary figure more completely explaining this process.]

$$cov(t_1, t_2) = \frac{\sum_i^M \left(\delta^{18}\mathrm{O}_i(t_1) - <\delta^{18}\mathrm{O}(t_1)>\right)\left(\delta^{18}\mathrm{O}_i(t_2) - <\delta^{18}\mathrm{O}(t_2)>\right)}{M} \tag{1}$$

Lu et al. (2023) used two benthic species of foraminifera to measure $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$. The $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ measurements were obtained from Uvigerina peregrina (U. peregrina) tests for the three shallowest cores (MC28A, MC26A, and MC25A), and from Cibicidoides wuellerstorfi (C. wuellerstorfi) tests for the other eight sediment cores. While C. wuellerstorfi calcifies at equilibrium, U. peregrina is isotopically heavier by 0.47‰ (Marchitto et al., 2014).

$\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ is dependent upon calcification temperature and the $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$, which is, in turn, controlled by temperature and salinity. Ultimately, increases in $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ are caused by decreasing temperature and/or salinification. In Equation 2, we show the experimental relationship for C. wuellerstorfi calculated by Marchitto et al. (2014), where $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ is in ‰PDB and $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$ is in ‰SMOW. This relationship can be applied to U. peregrina, as long as the 0.47‰ offset is removed.

$$\delta^{18}\mathrm{O}_{\mathrm{calcite}} = (-0.224 \pm 0.002)\frac{‰}{K}T + (3.53 \pm 0.02)‰ + (\delta^{18}\mathrm{O}_{\mathrm{seawater}} - 0.27‰) \tag{2}$$

In order to complete an inversion, the values reported from both C. wuellerstorfi and U. peregrina records must be put on a common scale. Effective modern-day $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ can be computed from the equilibrium value of $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$ and temperature from the inversion of Gebbie and Huybers (2010). These values can then be compared to the core-top values of $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$. We find that, after applying the experimental offset to the U. peregrina sediment cores, the measured $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$ values are variably offset from the computed effective $\delta^{18}\mathrm{O}_{\mathrm{calcite}}$. This variable offset,

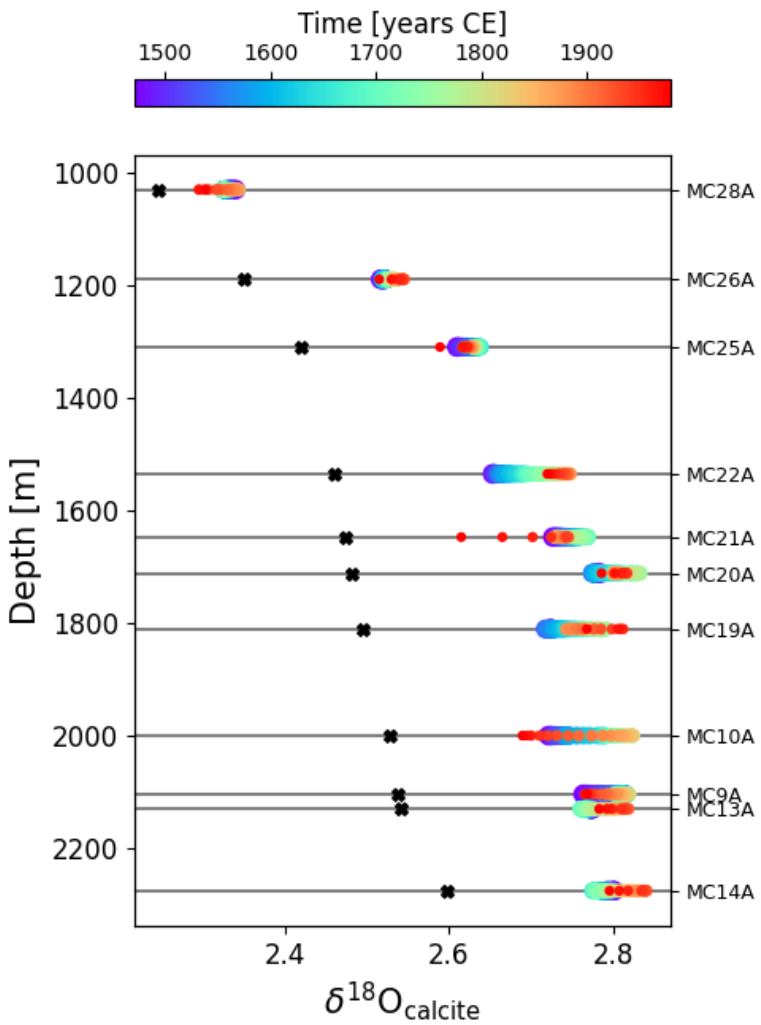

Figure 1. Rainbow dots, with size decreasing as time increases, show mean $\delta^{18}O_{calcite}$ for each sediment core, with 0.47‰ equilibration offset removed from U. peregrina records. Black crosses show effective $\delta^{18}O_{calcite}$ computed from the steady-state inversion of temperature and $\delta^{18}O_{seawater}$ from Gebbie and Huybers (2010) and observed at the sediment core sites. Computation of effective $\delta^{18}O_{calcite}$ is according to Equation 2.

2.2 Total Matrix Intercomparison (TMI), steady-state and transient versions

In order to relate signals at the sediment core sites to far-field ocean regions, an ocean circulation model must be applied. In Gebbie and Huybers (2010), a box inversion technique is used to compute the pathways connecting interior ocean grid boxes to the surface. The global ocean is decomposed into a $4°\times4°$ grid with 33 vertical levels, and the pathways that connect each interior grid box to the surface are computed by inverting six conservative tracers. This results in a matrix, $\mathbf{A}_{\mathrm{ss}}$, which contains the contribution of every surface grid cell to every interior cell, assuming that the ocean is at a steady-state. In Gebbie and Huybers (2012), this result is extended by incorporating radiocarbon observations in $\mathbf{A}_{\mathrm{t}}$, which contains the rate at which surface grid cells are transmitting water to each interior cell.

2.3 Coupled Climate Model Output

SST fields from historical realization (1850-2015CE) ‘r11i1p1f1‘ of the Community Earth System Model, version 2 (CESM2) (Danabasoglu et al., 2020) were used to constrain the relative variance of surface spatial modes.

The relationship between temperature and $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$ was obtained using output from the iCESM1.2 coupled climate model (Brady et al., 2019). The ‘TEMP‘ and ‘R18O‘ variables from the large ensemble pre-industrial fully coupled control run were observed at the grid cell closest to the mean sediment core position, and the linear least squares relationship of the two timeseries was found to be 10°C/‰.

3 Methods

3.1 Overview

This inverse problem is characterized by inferring some time-varying far-field properties $\mathbf{u}$ from data at the sediment core sites $\mathbf{y}$. Both $\mathbf{y}$ and $\mathbf{u}$ are vectorized arrays, with $\mathbf{y}$ varying by time t_y and sediment core c, and with $\mathbf{u}$ varying with time t_u, spatial mode m, and state variable (either temperature or $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$). The connection between the sediment core sites and any region of the ocean can be computed by applying the transient TMI matrix $\mathbf{A_t}$, which has dimensions $(b_s\times b_e)$, where b_s is the number of surface gridcells, and b_i is the number of interior gridcells. Each entry of $\mathbf{A_t}$ has units of mass flux, and represents the flux of water from the surface to the interior. Although this approach could be used to infer subsurface temperature and $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$, here we infer surface temperature and $\delta^{18}\mathrm{O}_{\mathrm{seawater}}$, as it can more readily be compared to other records.

We are concerned with solving for the time-varying evolution of some collection of far-field patterns. These patterns could be oceanographic regions or features, and could be surface or subsurface. In order to reduce the number of degrees of freedom in the problem, we solve for the magnitude of surface spatial modes, obtained from a singular value decomposition of source water maps (Section 3.2). These spatial modes can be propagated through the transient TMI matrix $\mathbf{A_t}$ and “observed” at the core sites, forming the model matrix $\mathbf{E}$ of spatial mode magnitudes at the core sites at lag times from an initial signal.

3.

$$\mathbf{y} = \mathbf{E}\mathbf{u} \tag{3}$$

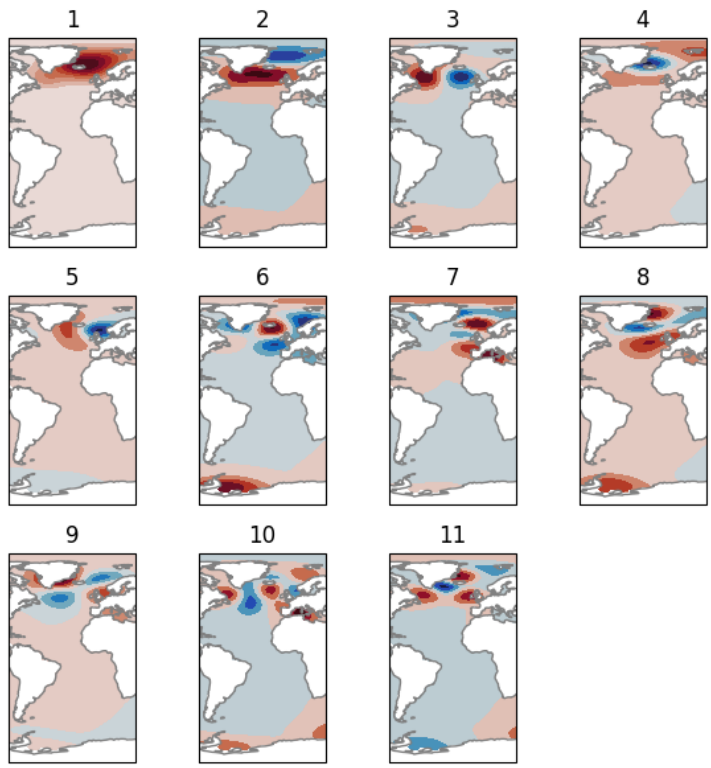

Figure 2. Surface spatial modes

3.2 Spatial Modes and Expected Variance

- How are spatial modes calculated? From the steady-state TMI matrix $\mathbf{A_{ss}}$, which is similarly dimensioned to $\mathbf{A_t}$, but has units of mass fractions, we can compute the amount of water at any interior ocean location
- Expected variance of spatial modes in CESM
- Relationship between temperature and δ^{18}O at core sites in iCESM

3.3 Computing Model Matrix

- Propagating modes through matrix
- Impulse response technique

3.4 Solving

- TWLS

3.5 Analysis techniques

- Surface reconstruction
- Reconstruction at core sites
- Cost functions

4 Results

4.1 Data Fit

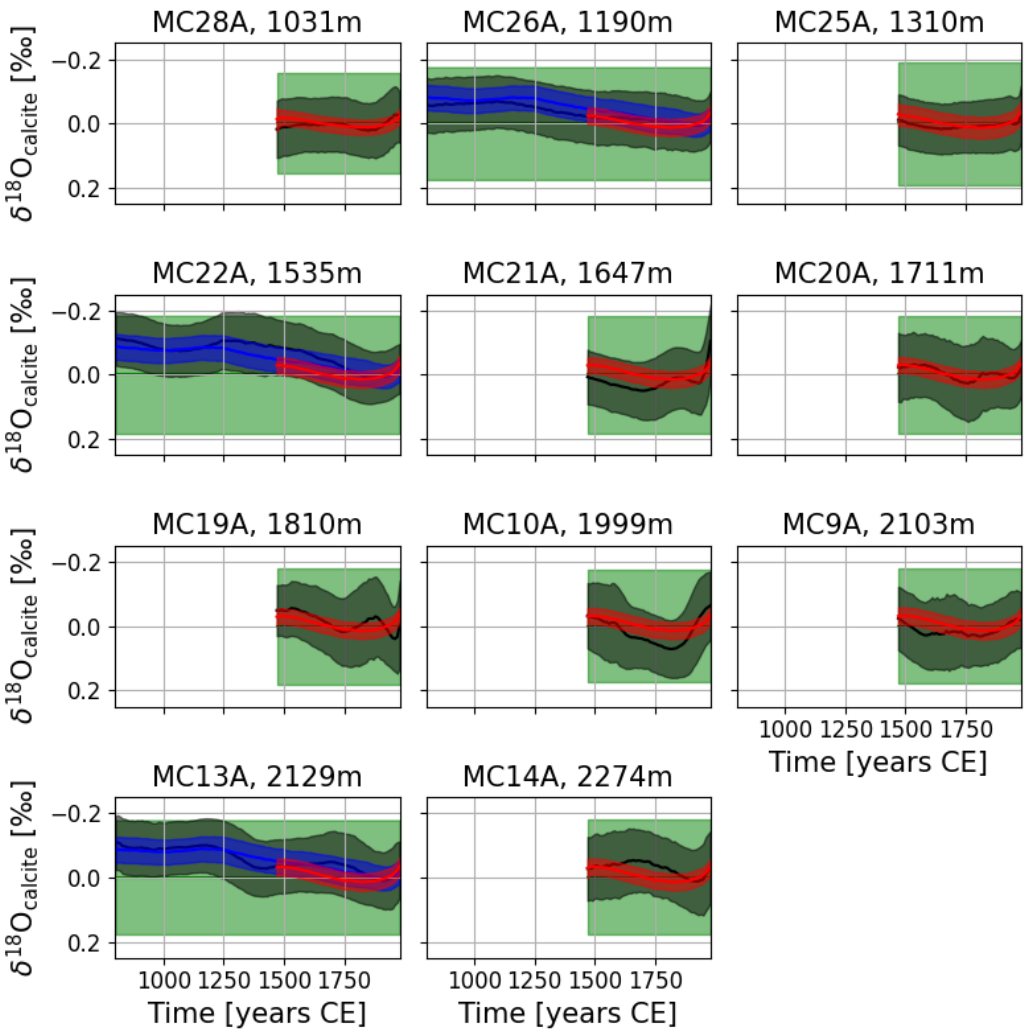

Figure 3. Reconstruction at core sites for all core estimate (red) and three oldest core estimate (blue). Black line shows sediment core record.

Statistics about how well the data is fit

4.2 Solution Comparison to Instrumental Record

4.3 Common Era Solution

5 Conclusion

Open Research Section

This section MUST contain a statement that describes where the data supporting the conclusions can be obtained. Data cannot be listed as "Available from authors" or stored solely in supporting information. Citations to archived data should be included in your reference list. Wiley will publish it as a separate section on the paper's page. Ex-

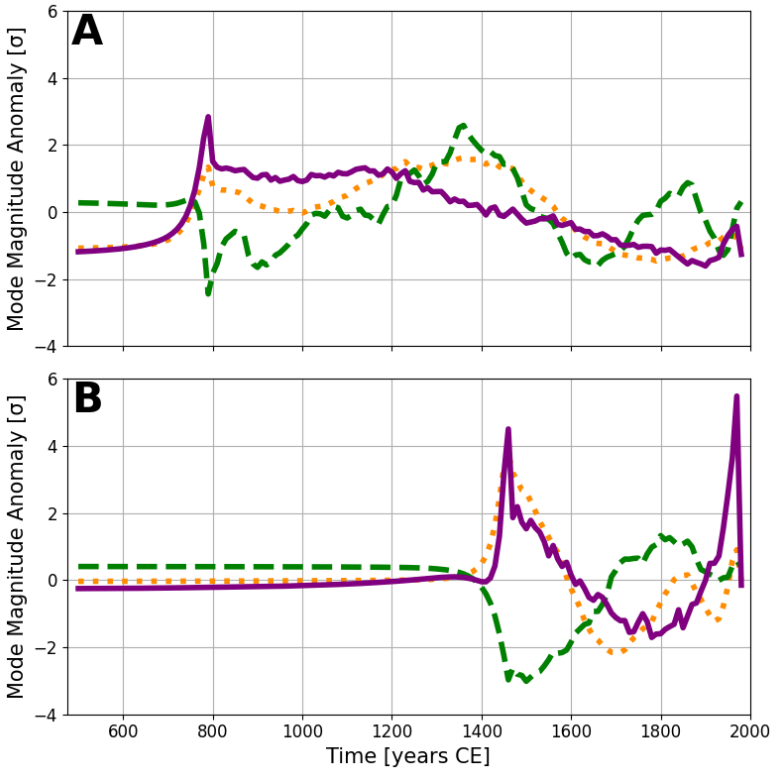

Figure 4. Temperature surface mode magnitudes for first three modes, mode 1 in purple/solid, mode 2 in green/dashed, mode 3 in orange/dotted for **(A)** estimate using three oldest cores and **(B)** estimate using all cores.

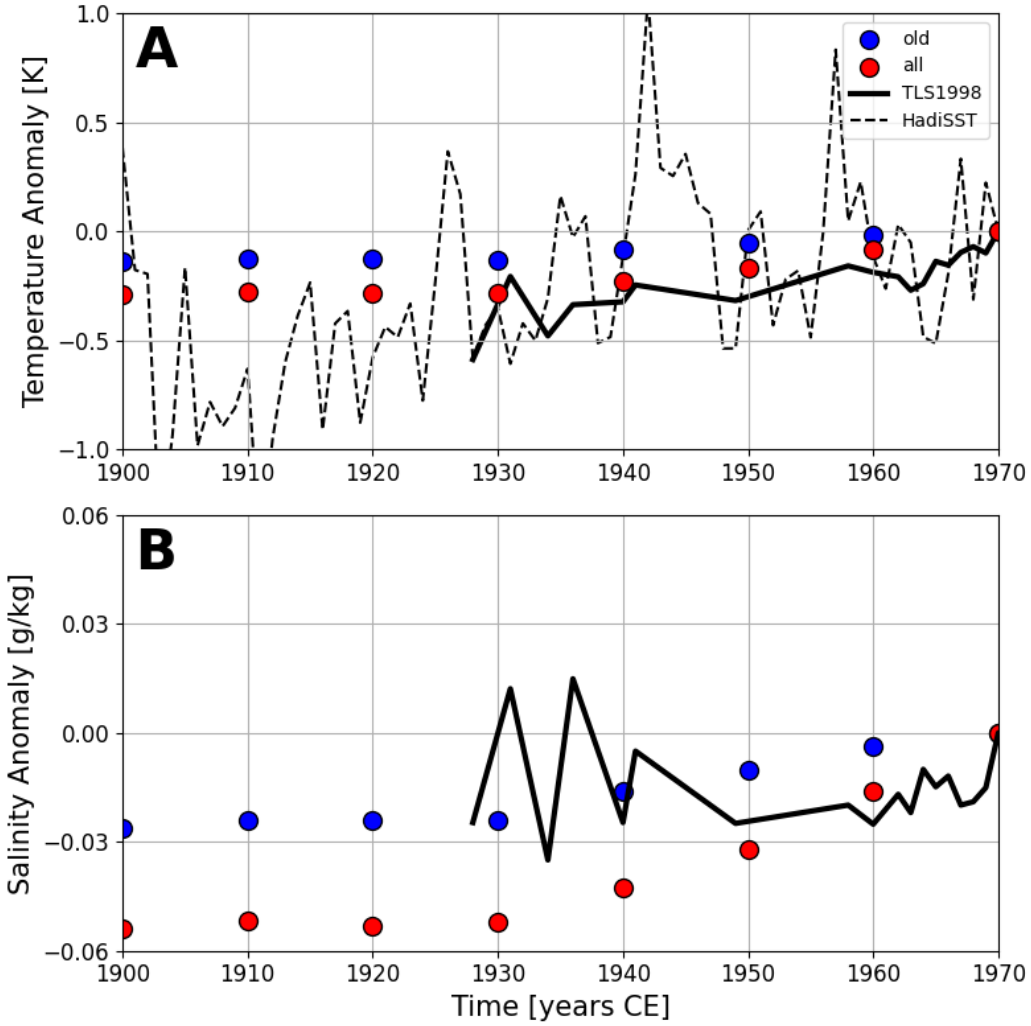

Figure 5. **(A)** Reconstructed Labrador Sea (60°W-30°E, 55°N-64°N) temperature and **(B)** salinity (computed as 0.5255 $\times$ $\delta^{18}O_{seawater}$) over the instrumental period (1850-1970) is shown for "old" (blue) and "all" (red) solutions. Solid line shows digitized Labrador Sea record from The Lab Sea Group, 1998 publication. Dashed line shows HadiSST Labrador Sea spatial mean (60°W-30°E, 55°N-64°N). All records are shown as an anomaly from 1970.

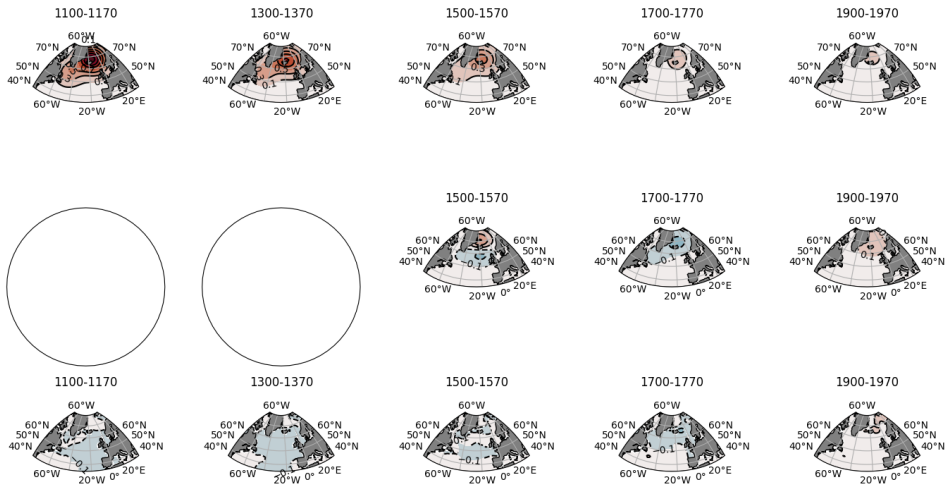

Figure 6. Reconstruction of surface potential temperature, 70-year mean, SVD approach, anomaly from 1850-1970. Top row is for old cores, middle row is for all cores, third row is Last Millenium Reanalysis SSTs (Hakim et al., 2016).

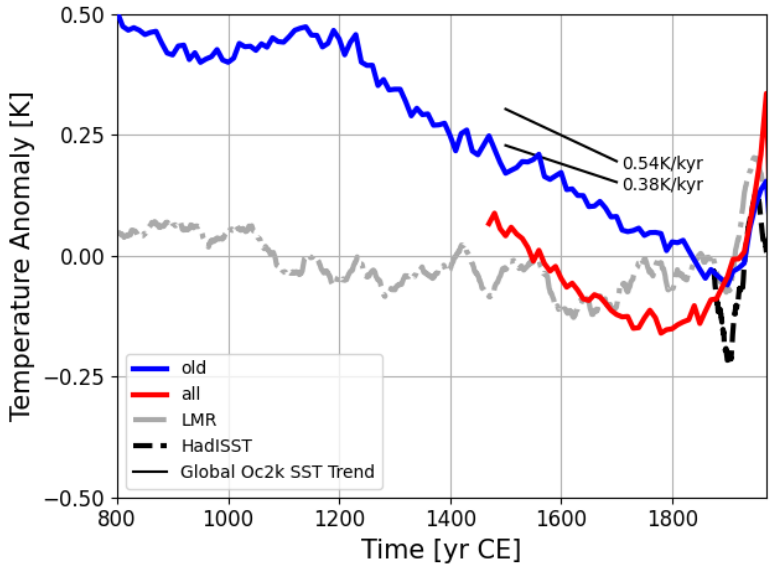

Figure 7. Mean North Atlantic temperature for "all" solution (red) and "old" (blue), Last Millenium Reanalysis (gray dash-dot), HadiSST (black dotted). Ocean2k trend of 0.54-0.38 °C/kyr (from Supplemental Figure 6) is shown in shaded gray region. All records are shown as anomaly from 1850-1970. For HadiSST and both reconstructions, the mean is computed over the region (50°W-20°E, 50°N-90°N)

[This will go in discussion, but for now: Estimated LIA N. Atl. cooling is calculated by computing the linear least squares slope between the maxima and minima occurring between 1000CE and 1900CE. Old: 0.67 ± 0.56 K/kyr, All: 0.77 ± 1.8 K/kyr]

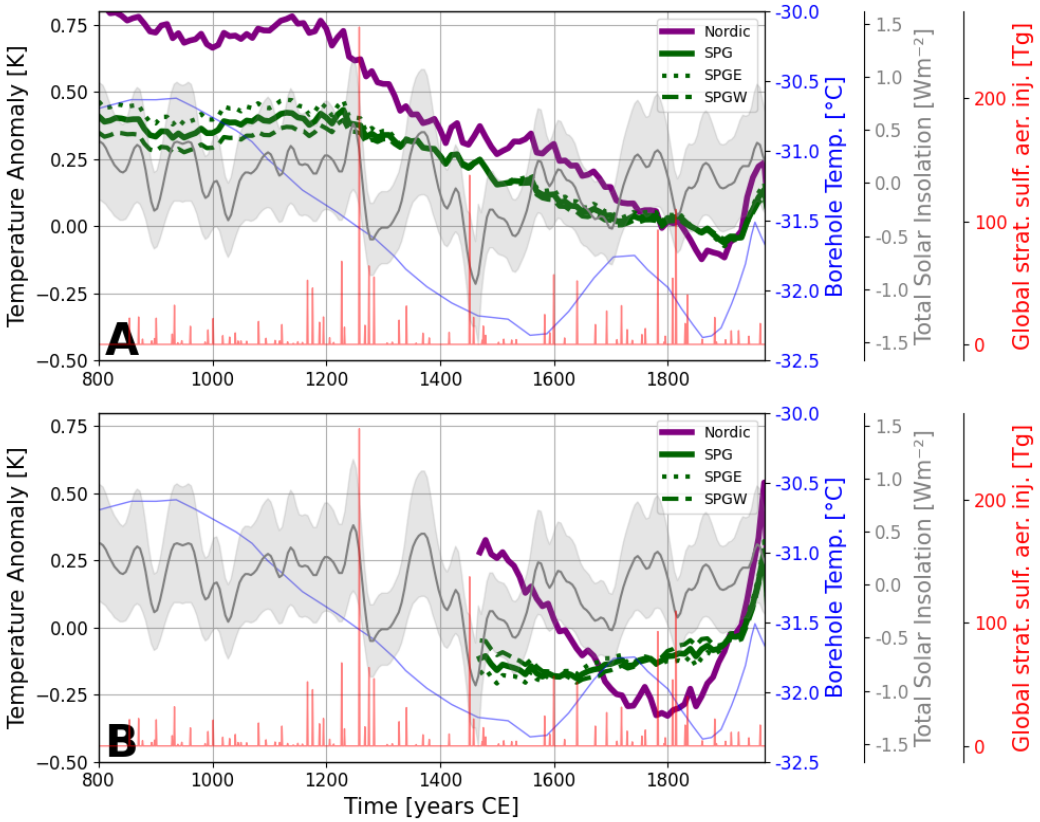

Figure 8. Temperature for **(A)** "old" and **(B)** "all" solutions observed for spatial averages in Nordic Seas (purple: 20°W-20°E, 65°N-80°N), the subpolar gyre (green: 60°W-0°, 50°N-65°N), the western subpolar gyre (green, dashed: 60°W-30°W, 50°N-65°N), and the eastern subpolar gyre (green, dotted: 30°W-0°, 50°N-65°N). All reconstructions are shown as anomaly from 1850-1970CE. Blue line shows digitized GRIP borehole record (Dahl-Jensen et al., 1998), grey line and shading show total solar insolation estimate with 1σ uncertainty (Steinhilber et al., 2009), and red lines show global stratospheric sulfate aerosol injection (Gao et al., 2008).
This figure is definitely too busy, I just thought it would be interesting to compare these records to some common forcing plot and see if anything pops out.
[For discussion: "old" solution reconstructs 0.9 ± 2.7K Nordic Sea cooling from 800-1860CE, and 0.5 ± 2.2K subpolar gyre cooling from 1230-1900CE.]

amples and complete information are here: https://www.agu.org/Publish with AGU/Publish/Author Resources/Data for Authors

Acknowledgments

Enter acknowledgments here. This section is to acknowledge funding, thank colleagues, enter any secondary affiliations, and so on.

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